Exp No:**2** Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Description: Use sample data set sales-data.csv.

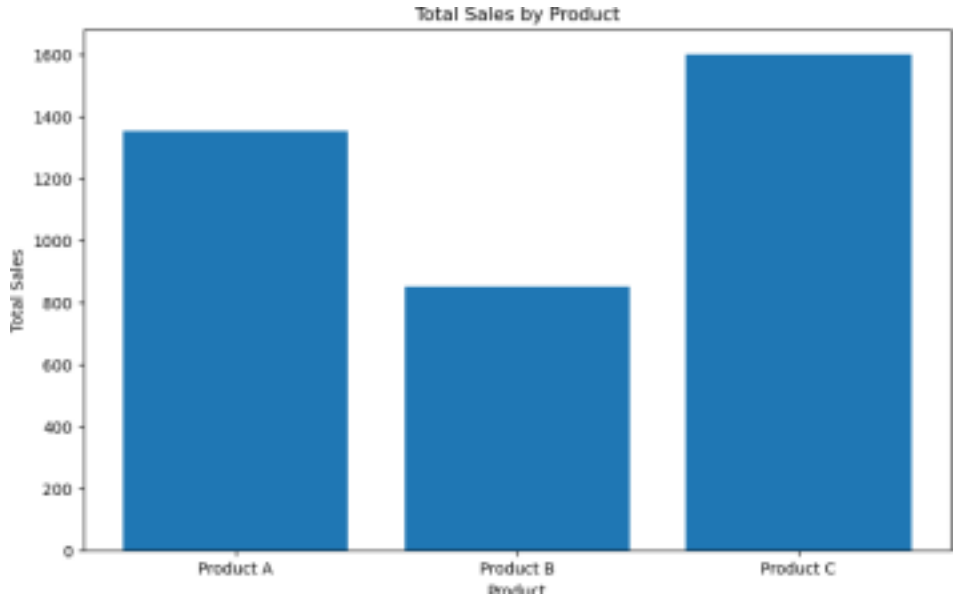
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Load the data into a pandas DataFrame
file_path='C:\sales_data.csv'
df = pd.read_csv(file_path)
# Display the first few rows of the DataFrame
print(df.head())
# Check for missing values
print(df.isnull().sum())
# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
# Summary statistics
print(df.describe())
# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
   'Sales': 'sum',
   'Quantity': 'sum'
}).reset_index()
print(product_summary)
```

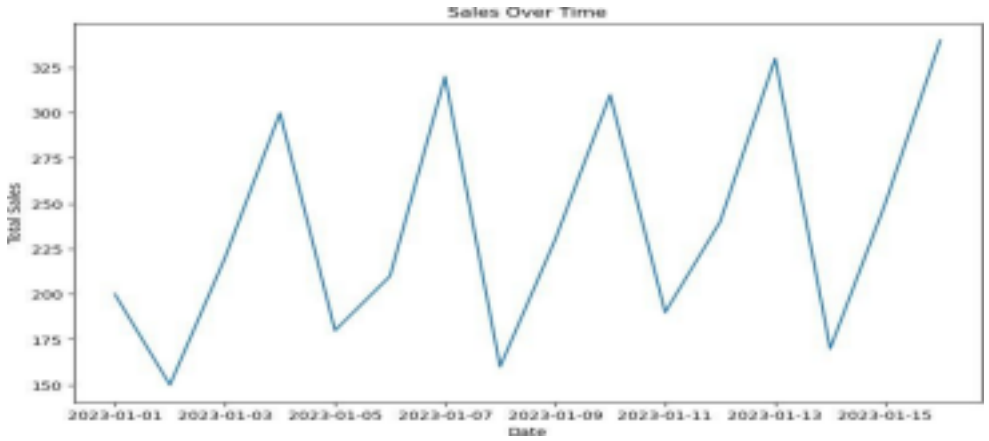
```
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'],
product_summary['Sales'])  plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales':
'sum'}).reset_index()  plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('SalesOver Time')
plt.show()
# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region',
columns='Product',  aggfunc=np.sum, fill_value=0)
print(pivot_table)
# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)
# Heatmap of the correlation matrix
import seaborn as sns
```

```
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True,
cmap='coolwarm')  plt.title('Correlation Matrix')
plt.show()
```


OUTPUT:

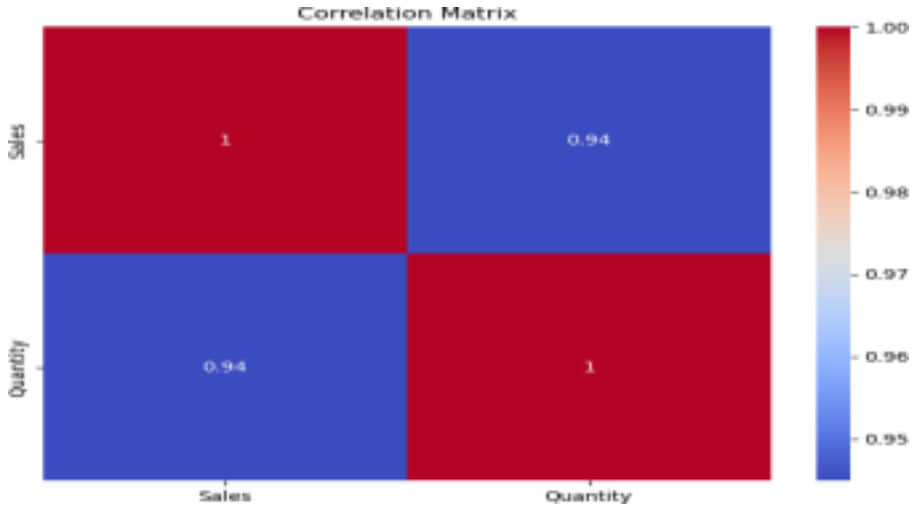
```
Date Product Sales Quantity Region
0 2023-01-01 Product A 200 4 North
1 2023-01-02 Product B 150 3 South
2 2023-01-03 Product A 220 5 North
3 2023-01-04 Product C 300 6 East
4 2023-01-05 Product B 180 4 West
Date 0
Product 0
Sales 0
Quantity 0
Region 0
dtype: int64
            Sales Quantity
count 16.000000 16.000000
mean 237.500000 5.375000
std 64.031242 1.746425
min 150.000000 3.000000
25% 187.500000 4.000000
50% 225.000000 5.500000
75% 302.500000 7.000000
max 340.000000 8.000000
     Product Sales Quantity
0 Product A 1350 33
1 Product B 850 17
2 Product C 1600 36
```






```
Product Product A Product B Product C
Region
East 0 0 1600
North 1350 0 0
South 0 480 0
West Sales 0          370       0.944922
           Sales      Quantity  0
           1.000000


Quantity 0.944922 1.000000
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: Futur
eWarning: The default value of numeric_only in DataFrame.corr is deprecated .
In a future version, it will default to False. Select only valid columns  or
specify the value of numeric_only to silence this warning.
  correlation_matrix = df.corr()
```


Correlation Matrix
Sales
Quantity
1
0.94
0.94
1
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95